# PAROLE Role 2 Dossier: TypeScript/Next.js Hero Experience, PC Workstation Guide & "Cybernetic Sentinel" Cockpit

STRICT STACK MANDATE: 100% TYPESCRIPT, NEXT.JS, TAILWIND CSS, FRAMER MOTION & RECHARTS (NO RUST!)

**ROLE 2 SPECIFICATIONS & NORTH-STAR REFERENCE BENCHMARKS**

- **Department:** Frontend Visual Experience, Home & About Hero Section, PC Workstation Onboarding & Real-Time Security Cockpit
- **Strict Technology Stack Mandate: 100% TypeScript, Next.js (App Router), Tailwind CSS, Lucide React Icons, Framer Motion & Recharts (Strictly NO Rust in the Frontend!)**
- **Assigned POWERHOUSE Skills:** `frontend-ui-dark-ts`, `design-philosophy`, `design-taste-frontend`
- **North-Star Reference Benchmarks ("Our Apple Standard"):**
  1. **Linear.app & Raycast.com:** Effortless onboarding, magnetic dark-mode polish, crystal-clear product storytelling, and immediate visual clarity on how a developer installs and uses the tool on their PC.
  2. **CrowdStrike Falcon + Vercel Geist Observability:** High-density 1px dark engineering panels (`#09090b`), zero visual clutter, and jitter-free `tabular-nums` monospace telemetry.

## 1. Your Mission in Plain, Human English

Hackathon judges do not read backend code first—**they experience your screen first.** Your mission is to make PAROLE look, feel, and behave like a $\$100$M Series B cybersecurity product built by the design teams at **Linear**, **Raycast**, and **Vercel**.

### Why We Use 100% TypeScript/Next.js (And Zero Rust) in the Frontend

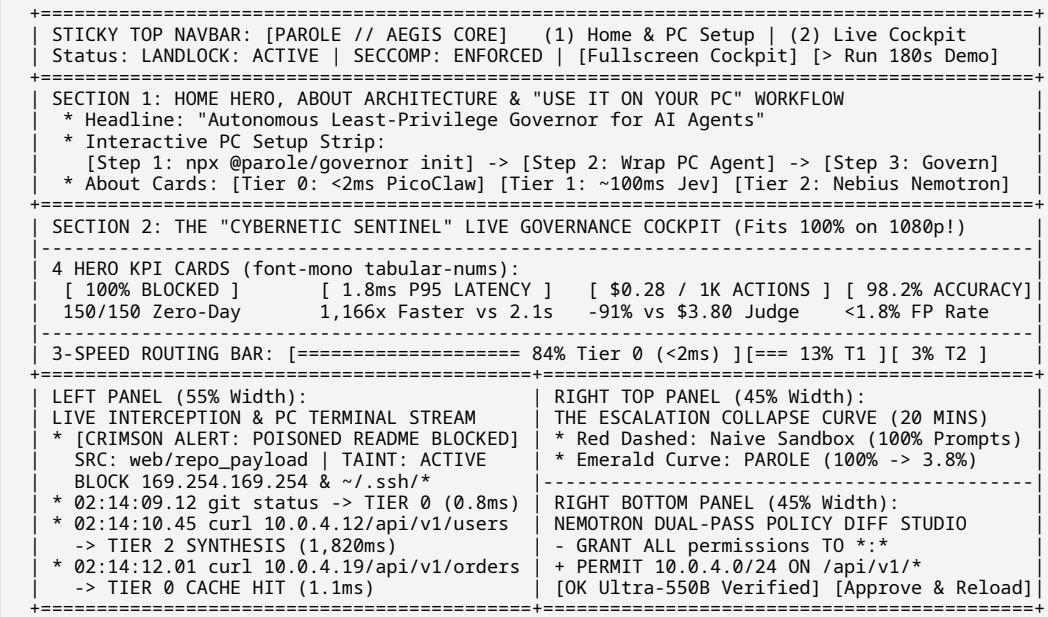
We want maximum visual fluidity, effortless usability, instant state transitions, and zero compilation headaches. By building $100$% in **TypeScript + Next.js + Tailwind CSS**, you get silky-smooth $60$ fps animations, interactive Recharts graphs, instant copy-to-clipboard terminal blocks, and a standalone Vercel deployment that loads in under $1$ second on any judge's PC.

### The Two-Part Experience You Are Building in One Unified Web App

Many developer security tools fail because they drop visitors straight into a confusing wall of logs with zero context. You will solve that by giving users a seamless **2-View Architecture** controlled from a sticky top navigation bar:

1. **View 1: The Executive Home, About Hero & "Use PAROLE on Your PC" Section:**
   - A breathtaking hero section that explains the **Permission Fatigue Trap** and our **3-Speed Sovereign Brain** in 10 seconds.
   - An interactive **"Use It on Your PC in 30 Seconds"** workstation guide showing how any developer runs PAROLE on their PC alongside OpenCode, Cursor, Claude Code, or OpenShell—complete with an interactive **PC Terminal Simulator** right in the browser!
2. **View 2: The "Cybernetic Sentinel" Live Governance Cockpit (With 1-Click Fullscreen Mode):**
   - The high-density command center where judges watch routine commands pass in Emerald Green ($1.1$ ms), see Poisoned README attacks get intercepted in Crimson Red (`[INJECTION: INTERCEPTED]`), watch the **Escalation Collapse Curve** drop from $100$% to $< 5$%, and click **"Approve & Hot-Reload"** on NVIDIA Nemotron policy diffs!

**UNIFIED 2-VIEW FRONTEND WIREFRAME (HERO + PC GUIDE + 1080P LIVE COCKPIT)**




## 2. Strict Visual Design System (“Anti-AI-Slop” Law)

Before prompting OpenCode or Antigravity, lock these visual rules into your project so the AI never generates cheesy hackathon cliches:

1. **Banned Visual Patterns (Never Allow These):**
   - **NO purple/pink/blue neon gradients** (`from-purple-500 to-pink-500` is strictly banned).
   - **NO emojis anywhere in the UI** (never use rocket, fire, or shield emojis—use sharp 16px `lucide-react` technical icons like `ShieldAlert`, `Terminal`, `Cpu`, `Lock`, `CheckCircle2`, `Laptop`, `Zap`).
   - **NO giant bubbly cards (`rounded-3xl`) or sloppy spacing.** Use crisp `rounded-md` corners and 1px borders (`border-zinc-800/80`).
2. **Exact Color Tokens (“Cybernetic Sentinel” Palette):**
   - **Primary Canvas:** Deep Carbon #09090b (`bg-zinc-950`)
   - **Elevated Surface:** Tactical Slate #121215 (`bg-zinc-900/80 backdrop-blur-md`)
   - **Hairline Borders:** Crisp Zinc #27272a (`border-zinc-800`)
   - **Tier 0 / Safe Resolution:** Emerald #10b981 (`text-emerald-400`, `bg-emerald-950/30`, `border-emerald-500/30`)
   - **Tier 1 / Fast Guardrail:** Cyan #06b6d4 (`text-cyan-400`, `bg-cyan-950/30`, `border-cyan-500/30`)
   - **Tier 2 / Nebius Synthesis:** Amber #f59e0b (`text-amber-400`, `bg-amber-950/30`, `border-amber-500/30`)
   - **Kernel Block / Injection Trap:** Crimson #ef4444 (`text-red-400`, `bg-red-950/50`, `border-red-500/50`)
3. **Typography Discipline:**
   - UI Headers & Prose: `Inter` or `Geist Sans` (`tracking-tight font-medium`).
   - All Numbers, Latencies, Dollar Costs, IPs & Commands: `JetBrains Mono` or `Geist Mono` with `font-mono tabular-nums` so numbers **never** wiggle horizontally when live counters update!

## 3. Your Complete Task List, Reverse-Engineering Loops & Verification Gates

### 3.1 TASK 1: Build the Professional Home, About Hero & “How to Use on PC” Section

Make sure any judge opening our Vercel link immediately understands what PAROLE is, why it matters, and how effortless it is to use on a personal computer or cloud workstation.

- **Subtask 1.1 — Sticky Top Navigation & Mode Switcher:** Build a sleek header (`h-14 border-b border-zinc-800 bg-zinc-950/90 backdrop-blur`) containing:
  - **Left:** Brand mark PAROLE + subtle badge `NEBIUS x NVIDIA 2026 // PROJECT AEGIS CORE`.
  - **Center Navigation Tabs:** (1) `Overview & PC Setup` (Scrolls smoothly to Hero & PC Guide), (2) `Live Governance Cockpit` (Locks into the full-viewport Dashboard), (3) `Architecture & Benchmarks v2.4` (Shows the 3-Speed & Red-Team specs).
  - **Right Action Controls:** Live status indicator `NEBIUS H100 + PICOCLAW: ONLINE` (pulsing emerald dot), Toggle button **`[ Fullscreen Cockpit Mode ]`** (Collapses the Hero section with a smooth Framer Motion transition so the Live Cockpit fills $100\%$ of a $1920 \times 1080$ screen for Role 3's video recording!), and Primary CTA button **`[ Trigger 180s Poisoned Repo Demo ]`**.
- **Subtask 1.2 — The High-Conviction Hero & "About PAROLE" Story:**
  - **Eyebrow Badge:** `AUTONOMOUS LEAST-PRIVILEGE GOVERNOR FOR AI AGENTS`
  - **Main Headline:** *"Stop Rubber-Stamping Sandbox Popups. Never Trust the Model That's Asking."*
  - **Subheadline:** *"Deny-by-default sandboxes bombard developers with confirmation prompts until they click 'Allow All'—exposing credentials to indirect prompt injection. PAROLE eliminates approval fatigue via a 3-Speed Sovereign Architecture: $< 2$ms local Go caching, $\sim 100$ms typed risk scoring, and dual-Nemotron policy synthesis on Nebius H100 GPUs."*
  - **3 Architectural Pillar Cards ("About the 3-Speed Brain"):**
    1. **Tier 0: PicoClaw Local Cache (<2ms | 84% Traffic):** Runs in `<10MB RAM` on your PC or VM, resolving known safe developer commands instantaneously at `$0.00` token cost.
    2. **Tier 1: System 1 Typed Risk Engine (~100ms | 13% Traffic):** TypeSafe Jev & Nemotron-Nano score blast radius and injection taint in a tenth of a second.
    3. **Tier 2: Nebius Cloud Dual-Nemotron (1.5s--3s | 3% Traffic):** `nemotron-3-super-120b` (`$0.30/M`) synthesizes surgical CIDR/path rules while `Nemotron-3-Ultra-550b` (`$1.00/M`) stress-tests boundaries before hot-reloading into Tier 0.
- **Subtask 1.3 — The Interactive "How to Use PAROLE on Your PC" Workstation Guide:** Build a dedicated, ultra-intuitive 3-step visual card strip titled **`USE PAROLE ON YOUR PC IN 60 SECONDS`** so judges see how easy it is for an everyday developer to use on Windows (WSL2), macOS, or Linux:
  - **Step 1 — Install the Lightweight PC Governor (<10MB RAM):** Show a clean terminal box with a working **1-Click Copy Button**: `npx @parole/governor init --agent=opencode --sandbox=nemoclaw`
  - **Step 2 — Code Normally Without Popup Fatigue:** Explain clearly: *"Keep using OpenCode, Cursor, Claude Code, or LangGraph on your PC. PAROLE sits out-of-process, auto-allowing $84$% of routine Git/test commands in $< 2$ ms while hard-blocking* `~/.ssh`*,* `~/.aws`*, and* `169.254.169.254`*."*
  - **Step 3 — Interactive "Try a Command From Your PC" Quick-Tester:** Right inside this PC section, give the user 4 quick-click preset buttons (using clean Lucide icons) so they can test how PAROLE behaves on a PC before even scrolling down:
    - `[ Run: git status (Tier 0 <2ms) ]`
    - `[ Run: pytest tests/ (Tier 0 <2ms) ]`
    - `[ Run: curl 10.0.4.12/api/v1/users (Tier 2 Nemotron Synth) ]`
    - `[ Simulate Poisoned README Attack (Kernel Block 0.3ms) ]`

    Clicking any preset immediately shows the instant PC terminal result **and** streams it into the Live Cockpit below!

**APPLE-STANDARD BENCHMARK TEST (TASK 1 — LINEAR.APP & RAYCAST STANDARD)**

**Copy-Paste Prompt for OpenCode / Antigravity / Grok Code:** *"Load* `frontend-ui-dark-ts`*,* `design-philosophy`*, and* `design-taste-frontend`*. Review our Hero, About Architecture, and 'How to Use on PC' sections against Linear.app and Raycast.com. Ensure the typography uses tight tracking, the copy-to-clipboard button shows a smooth emerald checkmark for 2 seconds, the 4 interactive PC command preset pills give instant visual feedback, and clicking 'Fullscreen Cockpit Mode' in the navbar smoothly collapses the Hero section so the dashboard fits $100$% within a single* `1080p` *viewport."*

**Re-Verification Gate 1:**

- ☐ Open `http://localhost:3000` → Confirm the Hero, About Architecture cards, and 3-step PC Guide look like a polished Silicon Valley developer tool (zero purple gradients, zero emojis).
- ☐ Click the copy button on `npx @parole/governor init` and click all 4 PC quick-test buttons → Confirm instant, smooth visual state updates.
- ☐ Click **`[ Fullscreen Cockpit Mode ]`** → Confirm the Hero collapses cleanly and the entire Live Cockpit fits on one screen without vertical scrolling!

## 3.2 TASK 2: Build the 4 Hero KPI Cards & 3-Speed Routing Distribution Bar

Anchor the top of the Live Cockpit with the exact verified numbers from Slides 7 & 8 of our Master Deck.

- **Subtask 2.1 — The 4 Canonical KPI Cards (`font-mono tabular-nums`):** Build a compact 4-column metric strip (`grid-cols-4 gap-3`):
  1. **ZERO-DAY INJECTION CONTAINMENT: `100%`** (`150 / 150` adversarial exfiltration commands blocked across `1,000` replay traces | Badge: `STATUS: VERIFIED`).
  2. **P95 END-TO-END LATENCY: `1.8ms`** (`1,166x FASTER` — reduced from `2,100ms` naive LLM-as-a-Judge baseline).
  3. **CLOUD COST / 1,000 ACTIONS: `$0.28`** (`-91% COST DROP` on Nebius vs. `$3.80` traditional GPT-4/Claude judge).
  4. **SAFE AUTO-GRANT ACCURACY: `98.2%`** (`<1.8%` False Positive rate on `850` benign developer tasks).
- **Subtask 2.2 — The 3-Tier Routing Efficiency Bar:** Directly below the 4 KPI cards, render a sleek segmented horizontal bar showing how traffic divides across the 3-Speed Brain:
  - **TIER 0 (<2ms PicoClaw Go Cache): 84%** (Emerald `#10b981`) — *Instant deterministic policy resolution & rule caching*
  - **TIER 1 (~95ms TypeSafe Jev / Nano): 13%** (Cyan `#06b6d4`) — *Lightweight speculative evaluation & local guardrails*
  - **TIER 2 (~1.9s Nebius Dual-Nemotron): 3%** (Amber `#f59e0b`) — *Full LLM reasoning & adversarial boundary verification*

**RE-VERIFICATION GATE 2**

☐ Verify every single number matches the Master Deck (`100%`, `1,000`, `<1.8%`, `98.2%`, `$0.28`, `$3.80`, `-91%`, `1.8ms`, `1,166x`, 84%, 13%, 3%).

### 3.3 TASK 3: Build the Live Monospace Interception Stream & Crimson Poisoned Repo Alert

This is the left-hand column (55% width) of the Cockpit where users watch real-time agent governance in action.

- **Subtask 3.1 — Real-Time Event Stream Table:** Each row in the live feed displays:
  - **Time & Node:** `02:14:08.104 | PC-LOCAL`
  - **Intercepted Command:** Monospace command string (`git status`, `curl -s http://169.254.169.254/...`, `curl https://10.0.4.12/api/v1/users`)
  - **Provenance & Taint Badge:** Trusted local flow `[SRC: TRUSTED_DEV]` (subtle zinc pill) vs. Untrusted repo/web flow `[SRC: web/repo_payload | TAINT_FLAG: ACTIVE]` (glowing crimson pill).
  - **Tier & Latency Pill:** `TIER 0 * 1.1ms` (Emerald), `TIER 1 * 94ms` (Cyan), `TIER 2 * 1,840ms` (Amber), or `SECCOMP BLOCK * 0.3ms` (Crimson).
  - **Interactive Row Inspection:** Clicking *any* row highlights it and loads its complete kernel telemetry (`$ sandboxing --mode=landlock`, `$ filtering --syscall=seccomp`, `[STATUS: ISOLATED_VM]`) and policy diff in the right-hand panel!
- **Subtask 3.2 — The "Poisoned README" Crimson Intercept Banner:** When Act 2 of the demo runs (or when a user triggers the Poisoned Repo trap), animate a high-contrast Crimson Alert Box at the top of the stream:
  - **Title:** `[INJECTION: INTERCEPTED] // OUT-OF-PROCESS KERNEL TRUST BOUNDARY`
  - **Subtitle:** *"An Agent Can Never Grant Permissions to Itself — Terminated by NemoClaw Landlock/seccomp"*
  - **Blocked Payload Details:** Shows `SRC: web/repo_payload`, `TAINT_FLAG: ACTIVE`, `BLOCK 169.254.169.254`, `BLOCK ~/.ssh/* & ~/.aws/*`, and `[OVERRIDE: FORBIDDEN]`.
- **Subtask 3.3 — Dual-Mode Bulletproof Engine (Live SSE + Built-In Client-Side Fallback): Non-Negotiable Winning Feature:** Connect your stream to Role 1's `http://localhost:8000/v1/policy/stream` SSE endpoint, **AND** build a complete deterministic TypeScript simulation engine directly inside Next.js (backed by Role 4's `telemetry_fallback_v2.4.json`). If the backend server is reachable, stream live from FastAPI. If a judge opens the public Vercel URL at 3:00 AM with no local Python server running, your built-in TypeScript engine seamlessly powers **"Trigger 180s Poisoned Repo Demo"**, **"Replay 1,000 Traces"**, and **"Approve & Hot-Reload"** with zero network errors!

**RE-VERIFICATION GATE 3**

☐ Stop the Python backend completely, open `http://localhost:3000`, and click **"Trigger 180s Poisoned Repo Demo"** → Confirm the entire 3-act choreography animates smoothly using the built-in TypeScript engine with zero console errors!
☐ Rapid-click the demo button 5 times in a row → Confirm button debouncing (`isDemoRunning`) prevents overlapping timers or duplicate events.

### 3.4 TASK 4: Build the Escalation Collapse Curve & Nemotron Dual-Pass Policy Diff Studio

This is the right-hand column (45% width) of the Cockpit, proving Slides 5, 6, and 8 visually.

- **Subtask 4.1 — The Escalation Collapse Curve Chart (`100% -> <5% in 20 Mins`):** Build an interactive Recharts area chart titled **`THE ESCALATION COLLAPSE CURVE (ADAPTIVE POLICY PROMOTION)`**:
  - **X-Axis:** Workflow Time (`0m`, `2m`, `5m`, `10m`, `15m`, `20m`). **Y-Axis:** Human Escalation / Prompt Rate (`0%` to `100%`).
  - **Curve 1 (Red Dashed Line — Naive Deny-by-Default Sandbox):** Stays stuck at `100% -> 95%` (causing human rubber-stamping).
  - **Curve 2 (Glowing Emerald Area Curve — PAROLE Reflex Engine):** Collapses from `100%` (cold start at `0m`) → 48% (`2m`) → 19% (`5m`) → `8.4%` (`10m`) → `4.9%` (`15m`) → **`3.8% (<5% at 20m)`**!
  - Add a quick tab switcher in this card header so the user can also toggle to the **Cloud Unit Economics Bar Chart** (`$0.28 PAROLE on Nebius` vs. `$3.80 Traditional LLM-as-a-Judge` = `-91% Cost Drop`).
- **Subtask 4.2 — The Nemotron Dual-Pass Policy Diff Studio:** Directly below the chart, build the **`REFLEX PIPELINE // NEMOTRON POLICY SYNTHESIS & BOUNDARY AUDIT`** panel:
  - **Model Header Pills:** `01 // PRIMARY REASONER: nemotron-3-super-120b-a12b ($0.30/M tokens)` and `02 // ADVERSARIAL RED-TEAMER: Nemotron-3-Ultra-550b-a55b ($1.00/M tokens)`.
  - **GitHub-Style Monospace Diff Box:**

    ```
    * - GRANT ALL permissions TO *:*  // [BLOCKED: Broad Human Rubber-Stamp]
    * + PERMIT 10.0.4.0/24 ON /api/v1/*  // [SYNTHESIZED: Strict CIDR/Path Scope]
    ```

  - **Adversarial Boundary Audit Stamp:** `[VERIFIED: ULTRA-550B BOUNDARY TEST PASSED]: Stress-tested against privilege escalation, metadata SSRF (169.254.169.254), and path traversal. Safe for Tier 0 promotion.`
  - **Interactive Action Button:** **`[ Approve & Hot-Reload into PicoClaw Tier 0 (<2ms) ]`**. When clicked: (1) Transitions to `[ Promoted to PicoClaw Cache (0.9ms) ]`, (2) Rule `+ PERMIT 10.0.4.0/24 ON /api/v1/*` appears in the **PicoClaw Active Cache Strip**, and (3) A new event `curl https://10.0.4.19/api/v1/orders` fires in the Left Live Stream and resolves in **Emerald Green (`TIER 0 * 0.9ms`)**!

**RE-VERIFICATION GATE 4 (FINAL FRONTEND SIGN-OFF)**

- ☐ Run `npm run build` → Confirm zero TypeScript or ESLint build errors.
- ☐ Deploy to Vercel and test the live URL at `1920x1080` and 125% zoom with Role 3 (Video Producer) → Confirm every font, chart tooltip, and diff box is razor-sharp for the 180-second video recording!